# BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2021

# CHEMISTRY

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

## GROUP – I

### Chemistry (Objective) Group - I

**Time Allowed:** 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** 7485

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. The end product of oxidation of acetylene is:

(A) Oxalic acid
(B) Glycol
(C) Glyoxal
(D) Formic acid

ایسٹیلین کی آکسیڈیشن کا آخری پروڈکٹ کونسا ہے؟

**(A)** آگزالک ایسڈ
**(B)** گلائیکول
**(C)** گلائی آکسل
**(D)** فارمک ایسڈ

Q2. Alkenes are prepared from alcohols by a process called:

(A) Dehydrogenation
(B) Dehalogenation
(C) Dehydration
(D) Dehydrohalogenation

الکینز کو الکحلز سے کس پروسس کے تحت تیار کیا جاتا ہے؟

**(A)** ڈی ہائیڈروجینیشن
**(B)** ڈی ہیلوجینیشن
**(C)** ڈی ہائیڈریشن
**(D)** ڈی ہائیڈرو ہیلوجینیشن

Q3. The functional group –COOH is found in:

(A) Carboxylic acids
(B) Aldehydes
(C) Alcohols
(D) Esters

فنکشنل گروپ –COOH کس میں پایا جاتا ہے؟

**(A)** کاربوکسیلک ایسڈز
**(B)** الڈیہائیڈز
**(C)** الکحلز
**(D)** ایسٹرز

Q4. What is the pOH of 0.02M $Ca(OH)_2$ solution?

(A) 1.698
(B) 1.397
(C) 12.31
(D) 12.61

$Ca(OH)_2$ کے 0.02M سلوشن کی pOH کیا ہے؟

Q5. At dynamic equilibrium:

(A) The reaction stops to proceed
(B) The amount of reactants and products are equal
(C) The speeds of the forward and reverse reactions are equal
(D) The reaction can no longer be reversed

ڈائنامک ایکوی لبریم کی حالت میں:

**(A)** ری ایکشن آگے بڑھنے سے رک جاتا ہے
**(B)** ری ایکٹنٹس اور پروڈکٹس کی مقدار برابر ہوتی ہے
**(C)** فارورڈ اور ریورس ری ایکشن کی رفتار برابر ہوتی ہے
**(D)** ری ایکشن مزید ریورس نہیں ہوتا

Q6. A reverse reaction is one that:

(A) Which proceeds from left to right
(B) In which reactants react to form products
(C) Which slows down gradually
(D) Which speeds up gradually

ریورس ری ایکشن وہ ہے:

**(A)** جو بائیں سے دائیں جانب ہوتا ہے
**(B)** جس میں ری ایکٹنٹس ری ایکٹ کر کے پروڈکٹس بناتے ہیں
**(C)** جو بتدریج آہستہ ہوتا ہے
**(D)** جو بتدریج تیز ہوتا ہے

Q7. Froth flotation process is used to concentrate the ore on:

(A) Density basis
(B) Concentration basis
(C) Wetting basis
(D) Magnetic basis

فراتھ فلوٹیشن میں اذر کو کنسنٹریٹ کیا جاتا ہے:

**(A)** ڈینسٹی کی بنیاد پر
**(B)** کنسنٹریشن کی بنیاد پر
**(C)** ویٹنگ کی بنیاد پر
**(D)** میگنیٹک کی بنیاد پر

Q8. In Solvay's process, slaked lime is used to:

(A) Prepare $CO_2$
(B) Prepare quick lime
(C) Recover ammonia
(D) Form $Na_2CO_3$

سالوے پروسس میں بجھے ہوئے چونے کو کس لیے استعمال کیا جاتا ہے؟

**(A)** $CO_2$ تیار کرنے کے لیے
**(B)** ان بجھا چونا تیار کرنے کے لیے
**(C)** امونیا حاصل کرنے کے لیے
**(D)** $Na_2CO_3$ بنانے کے لیے

Q9. Which disease causes liver inflammation?

(A) Typhoid
(B) Jaundice
(C) Cholera
(D) Hepatitis

کون سی بیماری جگر کی سوزش کا سبب بنتی ہے؟

**(A)** ٹائیفائیڈ
**(B)** یرقان
**(C)** ہیضہ
**(D)** ہیپاٹائٹس

Q10. Which gas protects the earth surface from ultraviolet radiations?

(A) $CO_2$
(B) CO
(C) $N_2$
(D) $O_3$

کون سی گیس زمین کی سطح کو الٹرا وائلٹ ریڈی ایشنز سے محفوظ رکھتی ہے؟

Q11. Infrared radiations emitted by the earth are absorbed by:

(A) $CO_2$ and $H_2O$
(B) $N_2$ and $O_2$
(C) $CO_2$ and $N_2$
(D) $O_2$ and $CO_2$

زمین سے خارج ہونے والی انفرا ریڈ ریڈی ایشنز کس میں جذب ہوتی ہیں؟

Q12. Glucose is:

(A) Hexahydroxy aldehyde
(B) Hexahydroxy ketone
(C) Pentahydroxy aldehyde
(D) Pentahydroxy ketone

گلوکوز ہے:

(A) ہیکسا ہائیڈرو آکسی الڈیہائیڈ
(B) ہیکسا ہائیڈرو آکسی کیٹون
(C) پینٹا ہائیڈرو آکسی الڈیہائیڈ
(D) پینٹا ہائیڈرو آکسی کیٹون

## Chemistry (Subjective) Group - I

**Time Allowed:** 01:45 Hours | **Maximum Marks:** 48 | **Paper Code:** 7485

Note: Part - I (حصہ اول)

Q2. (Marks 10) Write short answers to any FIVE parts.

(i) Give the characteristics of reversible reaction.

(ii) Derive equilibrium constant expression for the synthesis of ammonia from nitrogen and hydrogen.

(iii) Define dynamic equilibrium state.

(iv) How many times a solution of pH 1 will be stronger than that of solution having pH 2?

(v) What is auto-ionization?

(vi) Write the uses of nitric acid.

(vii) What is a ketonic group? Write the formula of acetone.

(viii) Write the condensed formula of n-butane and isobutane.

Q3. (Marks 10) Write short answers to any FIVE parts.

(i) Define substitution reaction.

(ii) Complete the reaction: $CH_3Br + 2[H] \xrightarrow{Zn/HCl}$

(iii) Why alkynes are also called acetylene?

(iv) Differentiate between essential and non essential amino acids.

(v) Write names of any two polysaccharides.

(vi) Why does acid rain damage buildings?

(vii) Why are the flood risks increasing?

(viii) Name any two secondary air pollutants.

Q4. (Marks 10) Write short answers to any FIVE parts.

(i) How detergents make the water harmful for aquatic life?

(ii) How temporary hardness is removed by Clarak's method?

(iii) Define soft and hard water.

(iv) Describe agricultural effluents.

(v) How $NaHCO_3$ is converted to $Na_2CO_3$.

(vi) How is ammonia recovered in the Solvay's process?

(vii) Describe the principle of Solvay's process.

(viii) Explain process of electro refining.

Note: Part - II, Attempt any TWO questions. Each question carries 09 marks.

حصہ دوم، کوئی سے دو سوالات کے جوابات تحریر کیجیے۔ ہر سوال کے 09 نمبر ہیں۔

Q5(a). (Marks 05) Explain Bronsted-Lowry concept of acids and bases giving at least one example.

Q5(b). (Marks 04) Give four uses of acetylene.

Q6(a). (Marks 05) Define functional group. Give examples of any four different oxygen containing functional groups.

Q6(b). (Marks 04) Explain oligosaccharides with examples.

Q7(a). (Marks 05) What is ozone? Write four effects of ozone depletion.

Q7(b). (Marks 04) Explain extensive hydrogen bonding ability of water.

## GROUP – II

### Chemistry (Objective) Group - II

**Time Allowed:** 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** 7488

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero

marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. Matte is a mixture of:

(A) FeS and CuO
(B) $Cu_2O$ and FeO
(C) $Cu_2S$ and FeS
(D) CuS and FeO

میٹ (matte) مکسچر ہے:

Q2. Formula of urea is:

(A) $NH_2COONH_4$
(B) $NH_2COONH_2$
(C) $NH_2CONH_4$
(D) $NH_2CONH_2$

یوریا کا فارمولا ہے:

Q3. A disease that causes bone and tooth damage is:

(A) Fluorosis
(B) Hepatitis
(C) Cholera
(D) Jaundice

ہڈیوں اور دانتوں کے خراب ہونے کی وجہ کون سی بیماری ہے؟

**(A)** فلوروسس
**(B)** ہیپاٹائٹس
**(C)** ہیضہ
**(D)** یرقان

Q4. Which is a secondary pollutant?

(A) $SO_2$
(B) $CO_2$
(C) $CH_4$
(D) $H_2SO_4$

کون سا سیکنڈری پلوٹنٹ ہے؟

**(A)** ڈائی سلفر ڈائی آکسائیڈ
**(B)** کاربن ڈائی آکسائیڈ
**(C)** میتھین
**(D)** سلفیورک ایسڈ

Q5. Formula of ozone is:

(A) $O_2$
(B) $O_3$
(C) O
(D) CO

اوزون کا فارمولا ہے:

Q6. Which is a disaccharide?

(A) Glucose
(B) Fructose
(C) Sucrose
(D) Starch

کون سا ڈائی سیکرائیڈ ہے؟

**(A)** گلوکوز
**(B)** فرکٹوز
**(C)** سکروز
**(D)** نشاستہ

Q7. The reduction of alkyl halides takes place in the presence of:

(A) Zn / HCl
(B) Na / HCl
(C) Mg / HCl
(D) Cu / HCl

الکائل ہیلائیڈز کی ریڈکشن کس کی موجودگی میں ہوتی ہے؟

Q8. Alkenes are prepared from alcohols by a process:

(A) Dehydrogenation
(B) Dehalogenation
(C) Dehydration
(D) Dehydrohalogenation

الکینز (alkenes) کو الکحلز سے کس پروسس کے تحت تیار کیا جاتا ہے؟

**(A)** ڈی ہائیڈروجینیشن
**(B)** ڈی ہیلوجینیشن
**(C)** ڈی ہائیڈریشن
**(D)** ڈی ہائیڈرو ہیلوجینیشن

Q9. General formula of alkyl radical is:

(A) $C_nH_{2n+2}$
(B) $C_nH_{2n-2}$
(C) $C_nH_{2n+1}$
(D) $C_nH_{2n}$

الکائل ریڈیکل کا جنرل فارمولا ہے:

Q10. Which is a Lewis base?

(A) $BF_3$
(B) $NH_3$
(C) $H^+$
(D) $AlCl_3$

کون سا لیوس بیس ہے؟

Q11. Who put forward Law of Mass Action?

(A) Newton
(B) Boyle
(C) Guldberg and Waage
(D) Charles

لا آف ماس ایکشن کس نے پیش کیا؟

**(A)** نیوٹن
**(B)** بوائل
**(C)** گلڈبرگ اور ویگ
**(D)** چارلس

Q12. Units of molar concentration are:

(A) mol $dm^3$
(B) mol $dm^{-3}$
(C) mol $cm^3$
(D) mol $cm^{-3}$

مولر کنسنٹریشن کے یونٹس ہیں:

## Chemistry (Subjective) Group - II

**Time Allowed:** 01:45 Hours | **Maximum Marks:** 48 | **Paper Code:** 7488

Note: Part - I (حصہ اول)

Q2. (Marks 10) Write short answers to any FIVE parts.

(i) How is dynamic equilibrium is established?

(ii) What do you mean by equilibrium constant?

(iii) What is active mass? Also write its units.

(iv) Why $H^+$ ion acts as Lewis acid?

(v) Which type of salt produce $SO_2$ gas on reacting with acid?

(vi) How salts are named? Give example.

(vii) How alkyl radical formed? Explain with example.

(viii) Define functional group with the help of an example.

Q3. (Marks 10) Write short answers to any FIVE parts.

(i) Why colour of bromine water discharges on addition of ethene in it?

(ii) Why hydrocarbons are soluble in organic solvents?

(iii) State the most significant difference between alkenes and alkynes.

(iv) Write open chain structures of glucose and fructose.

(v) Write chemical equation of photosynthesis.

(vi) Write chemical equation for ozone formation.

(vii) Define secondary pollutants and give an example.

(viii) Write temperature range of mesosphere.

Q4. (Marks 10) Write short answers to any FIVE parts.

(i) Point out two properties of water that make it an excellent solvent.

(ii) How decaying plants consume oxygen?

(iii) Write any two effects of water pollution.

(iv) What is fluorosis?

(v) Define metallurgy.

(vi) Name the various metallurgical operations.

(vii) What happens when ammonium carbamate is heated with steam?

(viii) How $NaHCO_3$ is converted to $Na_2CO_3$? Write its equation.

> Note: Part - II, Attempt any TWO questions. Each question carries 09 marks.

> حصہ دوم، کوئی سے دو سوالات کے جوابات تحریر کیجیے۔ ہر سوال کے 09 نمبر ہیں۔

Q5(a). (Marks 05) Write three uses of calcium hydroxide and two uses of magnesium hydroxide.

Q5(b). (Marks 04) Write any four uses of acetylene.

Q6(a). (Marks 05) What are alcohols? Draw structural formulae of primary, secondary and tertiary alcohols.

Q6(b). (Marks 04) What are polysaccharides? Give their properties.

Q7(a). (Marks 05) Give the characteristics of troposphere. Why temperature decreases upwards in this sphere?

Q7(b). (Marks 04) Define hard water. Write disadvantages of hard water.